

SLATE: System for Local Area Traffic Estimation: Theory and Application

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ABSTRACT

This research develops and applies a transportation planning model that integrates regional and local area forecasting approaches. While regional models have the scope to model the interaction of demand and congestion, they lack the spatial detail of a local approach. By combining the two methods, a number of important policy issues can be addressed, including the implications of traffic calming, changes in flow due to alternative traffic operation schemes, the influence of micro-scale zoning changes on nearby intersections, the impact of TDM on traffic congestion, and the consequences of a suburban light rail line.

INTRODUCTION

Macroscopic transportation planning models are traditionally used to estimate aggregate travel demand between areas in a metropolitan region and to estimate flows on major facilities. Regional level models are however not intended as a tool for determining the impact of individual sites on local traffic conditions, such as signalized intersections or levels of traffic on residential streets. Local area analysis often relies on traffic impact studies, which load new development onto an existing transportation network, and generally consider the effects of new traffic as simply additive to the old (1,2). At best, traffic impact studies will consider the effect of traffic on signalized intersections. Unlike the better regional transportation planning (3-5), they do not account for the feedback caused by changing local land use patterns and traffic flows on regional travel demand. For the very short term, a static analysis may be acceptable, but clearly congested conditions will result in altered travel patterns over the long term.

Planning and land regulation agencies would like to have the ability to forecast over a longer time period in small areas, a forecast which consider the interactions of travel demand and levels of congestion. An example of this can be found in Montgomery County, Maryland which has constructed regulatory systems around both a regional transportation planning model and local traffic impact studies (1,6). These two kinds of analysis, as might be expected, produce two different estimates of impacts and lead to varying interpretations of the consequences of new development. This desire for synthesis creates a need. The opportunity comes from sophisticated geographical information systems which provide road networks including each street, from cul-de-sac to interstate highway, and block and parcel level land uses enabling extremely detailed traffic zone systems and realistic network loadings. Coupling opportunity and need leads inexorably to the creation of a meso-level model, with more detail and feedbacks than a traffic impact study, but without the overhead or geographical scope of a regional transportation planning model. The meso-level model described in this paper has been titled SLATE, an abbreviation for System for Local Area Traffic Estimation.

A number of questions can be asked using a meso-level model such as SLATE that are not suited to either regional transportation planning models or traffic impact studies, including:

- What are the effects of residential traffic controls designed to reduce "through" trips in

neighborhoods, such as turn prohibitions, on major arterials and intersections?

- How much would the conversion of two-way signalized arterials to one-way streets reduce intersection congestion?
- How does widening (or narrowing) intersections, or constructing interchanges, change traffic patterns?
- How much does building near or far from a Metro (subway) station reduce trips?
- What is the effect of a surface light rail line on traffic routing?
- How much congestion relief can be expected from the implementation of a travel demand management program?

This paper describes a procedure to integrate a local area traffic model with a regional transportation planning model. Applications of this procedure for network design and land use regulation (including both the timing and zoning of land use) in several areas of Montgomery County, Maryland are presented.

DATA

Two data sources are used for land use in this analysis. For residential land uses on the ground, census block data is used for each census block in the study area. For commercial land uses and future residential, land use information from the tax assessors file is used. Each census block and commercial parcel is treated as a traffic zone in the study. By way of comparison the analysis of North Bethesda with SLATE uses 798 local zones and 20 external stations representing the rest of the region, while the regional modeling system (Travel/2) uses 19 traffic zones to represent this area, and 632 zones to represent the rest of the region.

The network used in the study area was developed from the Census TIGER files. Much work was still required to apply network attributes such as travel lanes, capacities, and freeflow speeds to the network. Each signalized intersection and all intersections of smaller streets with arterials are coded and modeled.

The SLATE model has been applied to several areas of Montgomery County. Three applications, Bethesda, North Bethesda, and Germantown West are described below. The regional model, Travel/2, provides inputs into the windowed local model, SLATE. Summary statistics for the models are provided in Table 1.

Table 1: Model Statistics

Area	Population	Jobs	Model	Regional Zones	Subzones
Bethesda	100,000	84,000	SLATE	30	911
North Bethesda	38,000	70,000	SLATE	19	818
Germantown West	35,000	9,000	SLATE	12	67
Montgomery County	750,000	400,000	Travel/2	292	-
Washington/Baltimore	6,900,000	3,700,000	Travel/2	651	-

note: 1993 estimates, ref. (6)

Travel/2 is applied to the combined Baltimore and Washington metropolitan areas, with a focus on Montgomery County, Maryland. The full region, home to over 6 million persons, 2 central cities and numerous suburban activity centers, is divided into 651 traffic zones for analysis. Thirteen of the zones serve as external stations to the region incorporating parts of four states, with access to the region from southern Pennsylvania, eastern West Virginia, central Virginia, and eastern, western, and southern Maryland. The plurality of traffic zones, however, are located in Montgomery County. The zone structure is derived from zones defined by the Baltimore Regional Council of Governments, the Metropolitan Washington Council of Governments, and the Montgomery County Planning Department for their transportation planning models.

MODEL

The SLATE model system is a combination of several attributes: its relationship with the regional model system, the specific model components, the interaction of those components, and the calibration of the model to better match observed data. Both the regional and windowed model are built upon the EMME/2 modeling platform (7), supplemented with additional programs written by the authors.

SLATE relies upon a regional transportation planning model, in this case Travel/2 (4) which models the Washington/Baltimore metropolitan region in the afternoon peak hour, to produce several specific inputs. The regional model computes the number of trips at stations external to the local window and the distribution pattern of trips between origins and destinations by work and non-work purpose. The data is tracked using a "traversals" methodology, which tracks the number of trips between any marked links. All of the links forming a cordon around the window, as well as all centroid connectors for traffic zones within the window were marked to obtain a submatrix conforming to the windowed model. Furthermore, two classes of trips, work and nonwork, were tracked separately.

While the regional model also produces a great deal of other data, that information can be more accurately estimated within the window. SLATE estimates the number of the trips by purpose, applies the regional distribution pattern to the specific trip generation in the area to obtain a trip table, assigns the vehicles to the complete road network within the window, and computes intersection delay. The reason for including every roadway in the study window is not to accurately forecast traffic on the smaller roads, but rather to improve traffic loadings on freeways, highways, arterials, and primary residential streets, as well as serving to flag potential problems

with neighborhood cut-through trips.

Trip Generation and Mode Factors

The land use is coded in extreme detail, with the maximum size of a traffic subzone being a census block, while some subzones represent individual parcels or buildings. To avoid confusion, traffic zones from the regional model are called zones, while those from the windowed model are called subzones, each zone is composed of one or more (usually around 30) subzones, with conforming boundaries. Over seventy land use activity categories are coded. Though the detail is probably excessive for the needs of the modeling exercise, it is already available through the County's geographic information system, so using it is as costly to the modelers as recoding the data to not use it.

Afternoon peak hour vehicle trip generation rates, specific to the land use activity, were derived from the ITE Trip Generation rates (8) and from local studies of major activity classes (residential, office, large retail) (1). They are shown in Table 1. The two way rate (R) multiplied by the origin share (So) gives the number of afternoon peak hour trip origins, R multiplied by (1 - So) gives the number of afternoon peak hour destinations.

Table 2: Trip Generation Rates

Land Use	Two-Way Rate (R)	Origin Share (So)	Units (U)	Purpose
single family house first 75 U	1.11 U	0.38	houses	residential
single family house above 75 U	0.82 U	0.38	houses	residential
townhouse first 100 U	0.83 U	0.41	houses	residential
townhouse above 100 U	0.48 U	0.41	houses	residential
garden apt first 75 U	0.48 U	0.41	houses	residential
garden apt above 75 U	0.47 U	0.41	houses	residential
high rise first 100 U	0.46 U	0.28	houses	residential
high rise above 100 U	0.34 U	0.38	houses	residential
convenience retail first 50000 sf	9.96 U	0.53	1000 sq.ft.	non-work
convenience ret. above 50000sf	7.43 U	0.53	1000 sq.ft.	non-work
regional stand alone retail	9.46U+ 0.2*(200-U)	0.53	1000 sq.ft.	non-work
regional mall	3.0 U	0.53	1000 sq.ft.	non-work
Wholesale	0.52 U	0.50	1000 sq.ft.	non-work
Hardware	4.87 U	0.54	1000 sq.ft.	non-work
Car Dealer	2.62 U	0.59	1000 sq.ft.	non-work
Gas Station	81.95 U	0.50	1000 sq.ft.	non-work
Other Automotive	2.87 U	0.54	1000 sq.ft.	non-work
Fast Food	34.32 U	0.45	1000 sq.ft.	non-work
Other Restaurant	7.66 U	0.30	1000 sq.ft.	non-work
Warehouse	0.74 U	0.65	1000 sq.ft.	non-work
Mini-Warehouse	0.26 U	0.48	1000 sq.ft.	non-work
Auto Repair	2.87 U	0.54	1000 sq.ft.	non-work
Health Service	4.08 U	0.70	1000 sq.ft.	non-work
Office first 25,000 sq.ft.	2.24 U	0.82	1000 sq.ft.	work
Office next 275,000 sq.ft.	0.92 U	0.82	1000 sq.ft.	work
Office above 300,000 sq.ft.	0.92 U	0.82	1000 sq.ft.	work
Industrial Park	0.75 U	0.47	1000 sq.ft.	work
Mixed Light Industrial	0.98 U	0.88	1000 sq.ft.	work
Government Campus & R&D	1.07 U	0.85	1000 sq.ft.	work

The number of trip origins (q_{is}) in a subzone is given by:

$$(1) q_{is} = R(S_o)Fx$$

and the number of trip destination (p_{jt}) by:

$$(2) p_{jt} = R(1 - S_o)Fx$$

where:

R = Trip Rate

So = Origin share in afternoon peak hour

Fx = transit adjustment factor

Though we are dealing with vehicle trips, changes in modal characteristics of transit, ride-sharing, and non-motorized transportation are reflected in several ways. First, vehicle trip rates are discounted based on distance of the site to the nearest Metro station, of which there are three in each study area.

The housing vehicle trip adjustment factor (Fh) is valid for sites within one-half mile of a Metro station entrance.

$$(3) Fh = 1 - 0.02 \frac{2640 - D}{100}$$

where:

D = distance in feet to nearest Metro station entrance.

The office vehicle trip adjustment factor (Fo) is valid within 1000 feet of a Metro station:

$$(4) Fo = 1 - 0.04 \frac{1000 - D}{100}$$

The retail vehicle trip adjustment factor, applied to sites with more than 300,000 square feet, is valid within one-fourth of a mile from a Metro station:

$$(5) Fr = 1 - 0.02 \frac{1320 - D}{100}$$

Second, explicit adjustments for travel demand management (TDM) programs are taken into account, which are used reduce trip rates under a given set of policies. The FHWA TDM model (9) is used for estimation. Third, any changes in regional mode choice is captured in the regional model, which produces the number of vehicle trips entering and leaving the window. Finally, any site which satisfies Montgomery County's Transit and Pedestrian Oriented Neighborhood Guidelines (10) had its vehicle trip rate reduced by 5%. The guidelines, which are rather stringent, require a high frequency of adjacent transit service and a neo-traditional street layout.

Non-residential trip ends were split into work or non-work trips as noted in table 1. Residential trip ends, as well as trips from and to external stations, were allocated in proportion to work and non-work non-residential trip ends.

$$(6) q_{ijrw} = q_{ijr} \left(\frac{\sum q_{ijnw}}{\sum q_{ijnw} + \sum q_{ijn}} \right)$$

$$(7) q_{ijrn} = q_{ijr} - q_{ijrw}$$

where:

q_{ijr} = origin residential trip ends (work and non-work)

q_{ijrw} = origin residential work trip ends

q_{ijnw} = origin non-residential work trip ends

q_{ijn} = origin non-residential non-work trip ends

This is done similarly for destination trip ends.

This method clearly skips over a number of complexities involved in chained trips, as well possible distinctions between trip length distributions among various non-work trip purposes. Obviously trips to the gas station are much shorter than trips to a regional shopping mall. These simplification were required by lack of data on various trip length distributions rather than for any theoretical reasons.

Destination Choice

Trip distribution determines the proportion of trips going to various destinations. The choice of destination within the regional model, factored into peak hour vehicle trips, is solved simultaneously with assignment to ensure consistency between travel times used to estimate the number of trips, and the travel times resulting from that level of traffic. In the regional model, an equilibrium assignment and a multimodal gravity model (11) are used to determine the distribution of trips by mode. The vehicle trip traversal submatrices (work and nonwork) from this equilibrium distribution/assignment are imported into the window.

The likelihood of a trip between two subzones in the window depends upon the number of trips between the regional zones (f_{ij}), and the number of trips generated in each subzone (q_{is} , p_{jt}).

Trip distribution as applied uses the doubly constrained gravity model structure:

$$(8) t_{ij} = k_{ij} p_{is} \frac{(q_{jt} f_{ij})}{\sum_i (q_{jt} f_{ij})}$$

where :

t_{ij} = the number of trips from regional origin zone i to regional destination zone j

p'_{is} = the normalized number of trips produced at origin: regional zone i, subzone s

q'_{jt} = the normalized number of trips attracted to destination: regional zone j, subzone t

k_{ij} = adjustment factor for zone interchange i to j, average = 1

f_{ij} = trips between regional zone i and zone j

$$(9) \sum q'_{jt} = \sum p'_{is}$$

The last line is the conservation constraint that the total number of trips originating in the window (including external stations) matches the number destined for the window. This is accomplished by splitting the difference between total trip origins and destinations, and adjusting the subzone totals accordingly.

$$(10) p'_{is} = \frac{p_{is}}{2} \left(1 + \frac{\sum q_{jt}}{\sum p_{is}} \right)$$

$$(11) q'_{jt} = \frac{q_{jt}}{2} \left(1 + \frac{\sum p_{is}}{\sum q_{jt}} \right)$$

Route Choice

The auto assignment is solved by the static user equilibrium method. The variables are freeflow travel time, volume, and capacity which are used to estimate congested travel time. The general form of the equation was developed in (12), and is a modified form of the standard BPR form, with an additional term to represent delay at volumes less than capacity. Link functions have to be developed considering intersection control. While the conventional model implicitly incorporate delay from intersection in link freeflow speeds and capacity, this model raises link capacity and freeflow speed on arterials from what would otherwise be expected in order to avoid double counting the additional time penalty at intersections.

$$(12) T_c = T_f \left(1 + A \exp\left(\frac{Q}{Q_0}\right) + B \left(\frac{Q}{Q_0}\right)^c \right)$$

where:

T_c = Congested Travel Time

T_f = Freeflow Travel Time (min) = $60L/V_f$

V_f = Freeflow Travel Speed (mph)

L = Length (miles)

Q = flow (in vehicles per hour per lane)

Q_0 = capacity (in vehicles per hour per lane)

A, B, c = estimated parameters

Table 3: Volume-Delay Function Parameters

Route Type	A	B	c	Q ₀	V _f
Interstate	0	1	4	2000	55
Ramp	0	1	4	1200	45
Major Arterial	0.255	0.30	4	1600	45
Minor Arterial	0.188	0.48	4	800	35
Residential	0.111	0.68	4	400	10

Intersection Control

Most transportation planning models either ignore traffic signal timing or consider it exogenous to the model system, and thus independent of route choice. This presumption of independence is counter-intuitive, and is more a result of modeling difficulties (such as increased data and computational needs, as well difficulties in modeling the non-convex traffic assignment (13) than any theory. But empirical evidence (14) suggests that signal delay, including both stopped and approach delay, amounts to over 29% of total time on arterials in Montgomery County, and the vast majority of all delay.

Several attempts have been made to integrate an assignment algorithm with intersection control (15-17) in more or less realistic systems. The nature of combining the two systems is such that initial assumptions and the equilibration algorithm can influence final outcome. Because the SLATE model aims to reproduce realistic flows, rather than optimize signal settings, assumptions which seemed realistic were taken, rather than spending considerable effort searching for optimal settings. It should also be noted that an isolated intersection model is used for convenience, though it is known that signal settings are coordinated in the windowed model area.

The output of the intersection control model is the green time, cycle length, and effective number of turning lanes (when lanes are shared) for a turning movement. The cycle time and green time are estimated using methodologies suggested by Webster (18), while lane adjustment factors and lane utilization factors are adopted from Chapter 9 of the Highway Capacity Manual (19). The green time is assigned to equalize the volume/saturation flow on the critical approaches.

$$(13) C = \frac{1.5L+5}{1.0 - \sum_{p=1}^4 \frac{Q}{Q_0}}$$

where:

C = Cycle Length (sec)

L = Lost Time per Cycle (sec) = 4

Q = Flow on Movement in Vehicles per study period (T)

Q₀ = Saturation Flow Rate (2000 Vehicles per Hour of Green)

p = phase

This in turn is imported into the assignment model, where a turn penalty function estimates delay model based on the model below (20):

$$(14) d = \frac{C}{2} \left(1 - \frac{g}{C}\right) + \frac{T}{2} \left(\frac{Q}{Q_0 \frac{g}{C}} - 1\right)$$

where:

d = average delay (sec)

T = Length of Congested Time Period (sec) = 3600 sec (1 hour)

g = green phase length (sec)

To ensure stability and more rapid convergence, an equilibration algorithm is used. On the initial iteration, before link volumes are known, the green time (g_0) is allocated based upon the approach capacities, the approaches with the most capacity is given the most green time. On subsequent iterations, the green time (g_i) is calculated using the volumes from the previous iteration in a manner which would equalize the volume to capacity ratios for the critical movements at each approach assuming a four phase intersection (or three phase for a "T" intersection). Excess time from the non-critical left turn movements are given to the complementary through movement.

On the subsequent iterations however to ensure a convergence, an accumulating average is used to estimate the green time actually input into the model for the next iteration of assignment. This value is the adjusted green time (g'_i), which is computed using the following equation:

$$(15) \quad g'_i = g'_{i-1} \left(\frac{i-1}{i}\right) + g_i \left(\frac{1}{i}\right)$$

Calibration

Despite the best efforts to capture detail in the model specification, no model will perfectly replicate on-ground conditions. Model calibration was conducted in several stages, principally to match on-ground traffic counts in the base year (1990 for Bethesda and Germantown West, 1992 for North Bethesda). The first two model calibrations are internal to the model, the last is a post-processor.

A general issue surrounding model calibration is how to adjust the model estimates to match the observed. Two approaches come immediately to mind, adding the difference (observed - estimated in the base year) to the forecast estimate or multiplying the forecast estimate by the ratio of observed to estimated in the base year. As can be seen from a simple numerical example, neither is ideal under all circumstance. Suppose in the base year, the model predicts 10 cars, the observed value is 5 cars, using a ratio, we would multiply the model forecast estimate by 0.5, we are off by 100%. However if traffic in the future is of a different order of magnitude, for instance an office building is added adjacent to the location of the count, adding 200 vehicles to the link, then we would adjust downward to 100 vehicles if we applied the multiplicative correction. In fact, the count may be incorrect because of an error of 5 vehicles being added from a "ghost" source that appears in the data but is vacant (temporarily or permanently) on the ground. The error is in fact a 5 vehicle error, not a 100% error. On the other hand, if the link to begin with has 1000 vehicles, and the model predicts 1100, we either have a 10% error or a 100 vehicle error. If in the forecast,

we expect 2200 vehicles before correction, should we correct to 2000 or to 2100 vehicles? If the error is, as before, because of missing or double-counted generators, then the additive approach is correct, but if it is because of a misspecification of trip generation rates, then the multiplicative adjustment is more appropriate. Since we do not know a priori the source of error (if we did, we would have corrected it elsewhere), then we need to make a judgment. The approach taken here is to combine the two methods:

$$(16) X' = WX\left(\frac{X_o}{X_e}\right) + (1 - W)(X + (X_o - X_e))$$

where:

X' = adjusted forecast estimate

X = unadjusted forecast estimate

X_o = base year observation

X_e = base year unadjusted estimate

W = weight to combine methods (between 0 and 1), $W = 0.5$

First, the number of trips entering or leaving the window at external stations was adjusted. This corrects for any under or over estimation from the regional model. Cordon counts from several sources were used to provide a base year observed value.

Second, a gradient adjustment method (21) was applied to adjust the window's modeled base year vehicle trip table to minimize difference between modeled and observed link traffic counts. There are a number of reasons to expect a mismatch, including errors in specification of trip generation rates, regional trip distribution, or link or intersection delay functions. Running the algorithm for 10 iterations in the Bethesda/Chevy Chase window cut the error by two thirds. The r -squared (estimated traffic count regressed on observed traffic count) increased from 0.77 before calibration to 0.90 after iteration 10. The entire trip table was reduced from 58,500 to 54,300 trips. The improved trip table was used to adjustment matrices. For the base year, this returns the improved trip table, for future year it provides a more probable estimate.

Third, as a post-processor, turning movements at individual intersections from the model were adjusted to match turning movement counts. The adjusted turning movement counts were used to calculate critical lane volumes and other level of service indicators used to recommend policy.

Convergence

A key question to ask of the model is "Does the model converge to a single equilibrium answer"? A related question is "Does reality converge to equilibrium"? Depending upon our answer to the second, the desired answer to the first becomes clearer. The model comes close to its final answer after the fifth iteration, but because we are dealing with a "non-convex" problem combining signal timing and route choice, each with different objective functions and solution methods, there is no reason to expect a final true equilibrium answer. This disequilibrium is despite the equilibration algorithm used in the intersection control module. In fact, even at iteration 15, there are small fluctuations (plus or minus 3 vehicles per hour on the average link) of volume

(flow) on links from iteration to iteration, although fluctuations are tending to get smaller iteration after iteration. For this reason, interpretation of the model for comparison of alternatives should clearly be looking at larger differences than this to ensure that they are significant. Because the average volume difference is usually positive, this indicates that most links are gaining volume from iteration to iteration. This is also to be expected: on the first iteration, all cars are assigned to the shortest (freeflow time) path; but after they all take this path, it is no longer the shortest. On subsequent iterations the equilibrium assignment moves vehicles to less congested routes. Because there are more routes which are not the shortest path than there are routes which are the shortest path, most routes gain volume as the traffic is spread over the network.

Table 4: Model Convergence Results

Iteration	Average Volume Difference	Average Absolute Difference
1	5.74	19.14
2	5.58	20.57
3	1.77	7.08
4	1.73	6.59
5	.81	11.07
6	.44	5.75
7	.09	5.53
8	.31	5.21
9	.09	3.16
10	-.06	.78
11	-.46	3.58
12	.12	1.67
13	.00	2.27
14	.05	3.06

note: from Bethesda CBD final (approved) scenario (rf9630)

APPLICATION

The SLATE model has been applied in several areas of Montgomery County: Germantown West, Bethesda, North Bethesda, Rockville, and Glenmont. The principal intention of these applications was to conduct a comprehensive local area transportation review (CLATR), which would supplant developer funded traffic impact studies in these areas. All of the areas are nodes of growth designated by the County's master plans, most are centered on Metro stations. The modeling analysis was coupled with various policy changes, urban design recommendations, and incentives and requirements for use of travel modes other than the single occupant vehicle.

To illustrate the use of the model for linking transportation and growth management, the case of Germantown West, the first application of SLATE, will be examined. Germantown West is the western half of a "corridor city", an area planned for growth along the I-270 corridor which bisects Montgomery County (Germantown MP). Within this area is a smaller area, the Germantown Town Center, which is planned to have intensive office development next to a

MARC commuter rail station. Because, once all of approved development was constructed, the expected amount of traffic congestion was in excess of the area's level of service standard, the area had been placed in moratorium for new development approvals. This temporary moratorium was in contradiction to the County's plan for the area, the policy issue was raised about what would be required, in terms of policy changes, transit improvements, or highway improvements, to accommodate the completion of the Germantown Town Center development under two regulatory tests: policy area transportation review - using the regional model and considering interaction effects, and local area transportation review - basically a traffic impact study on the effects of new development on the level of service of nearby links and intersections. The SLATE model was used to develop highway and intersection improvement recommendations, and to test the impact of transit improvements (coupled with Travel/2 and changes in mode shares), and to find new level of service standards which would make the "problem" go away.

Staff recommended making a number of highway and intersection improvements to meet existing level of service standards, after some modifications to meet financial constraints, and intensive lobbying by local civic groups, construction was begun on the key recommended new road (MD 118 relocated), and development capacity was allocated to the Germantown Town Center. Table 5 shows the level of service under four main scenarios tested with SLATE (though numerous other alternatives were considered). Base development is defined as that which is on the ground, the pipeline is that which is approved but unbuilt, the full development of the town center was an additional 9,000 jobs and 2,000 housing units, and the approved transportation network is that which is built plus all improvements which are funded for completion within four years, which for the Germantown West SLATE analysis assumed completion by 1997.

Table 5: Summary of Intersection Level of Service, Germantown West SLATE

Intersection	1990(Base); Existing Network	Base + Pipeline ; Approved Network	Base + Pipeline + Town Center; Approved Network	Base + Pipeline + Town Center; Recommended Network
MD 118/Aircraft Drive	E	D	D	D
MD 118/Crystal Rock Drive	C	E*	E*	D
MD 118/ Middlebrook Road	C	D	E*	D
MD 118 N. /Wisteria Drive	A	D	D	D
MD 118 S./Wisteria Drive	A	D	E	D
Old MD 118/MD 117	E	F*	F*	C
Relocated MD 118/MD 117	-	-	-	D
MD 117/Waring Station Road	D	D	D	C
MD 117/Matney Road	D	D	D	D
MD 117/Great Seneca Hwy.	E	E	E*	E
Great Seneca Hwy./Matney Road	-	F*	F*	E
Great Seneca Hwy./ Wisteria Drive	D	E	E*	E*/**
Great Seneca Hwy./Middlebrook Road	C	E	E	E
Crystal Rock Drive/Wisteria Drive	-	E*	F*	E*/**
Crystal Rock Drive/Middlebrook Road	B	E*	F*	D
Crystal Rock Drive/Aircraft Drive	A	D	D	D
Crystal Rock Drive/Father Hurley Blvd.	A	D	D	D
Waters Landing Drive/Father Hurley Blvd.	A	D	D	D
Middlebrook Road/Waring Station Road	A	E*	F*	E
Wisteria Drive/Waring Station Road	B	D	D	C

note: * intersections with unacceptable critical lane volume > 1525, ** with recommended intersection improvements, intersections operate at acceptable level of service (CLV ≤ 1525); source: Table 2, Germantown Town Center Staging Analysis, MCPD 1992

CONCLUSIONS

The SLATE model provides a useful tool situated between the traditional traffic impact study and the comprehensive regional travel demand analysis. By combining computerization, which enables the flexible estimations of impacts of scenarios while considering regional demand impacts, with the detail of including every street and assigning a traffic zone to each parcel, the micro-scale planning needed for studying intersections in tightly packed urban areas can be conducted.

This model shows the sensitivity of traffic patterns to intersection control. Also worth noting are the air quality impacts of stopped delay and running speed. Given current fuel choices by the vehicle fleet and present technologies, valid estimates of air pollution need to be able to determine stopped delay, running speed, and total traffic demand. Incorporating the intersection in the planning model is necessary for proper implementation of Clean Air Act requirements

While the model is useful, it can always be improved. A key theoretical weakness is the assumption of isolated intersection control. A more rigorous approach would optimize signals on a system wide basis as with TRANSYT, or on an arterial basis such as MAXBAND. Integration with a local area network simulation, such as NETSIM, may also provide more accurate intersection flows. It should be noted that all of these programs still require exogenous specification of demand.

A second direction for improvement would be the inclusion of transit modeling within the SLATE framework. The network detail in SLATE is finally sufficient for accurate modeling of transit routes and access to transit, a difficult area to model using a coarse regional system. This would also require generalizing trip rates from vehicle trip rates which are adjusted to person trip rates to which mode choice can be directly applied.

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Figure 1: Flow Chart of SLATE Model

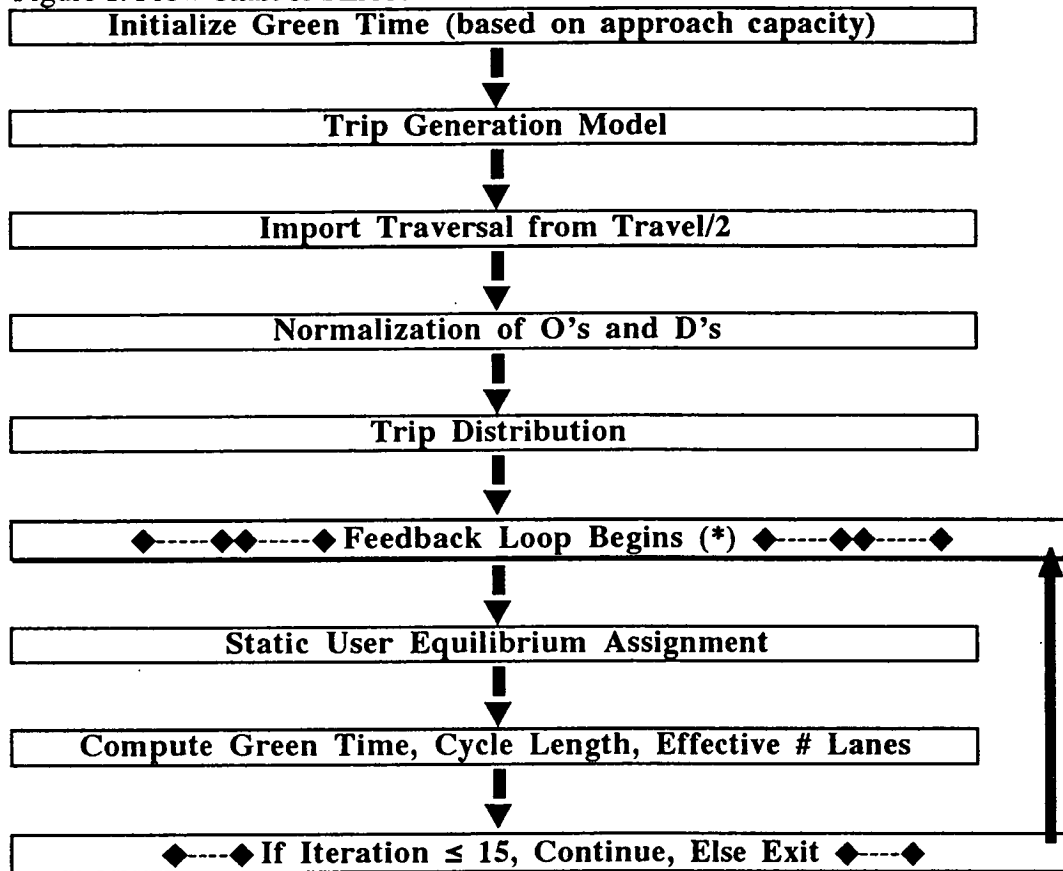
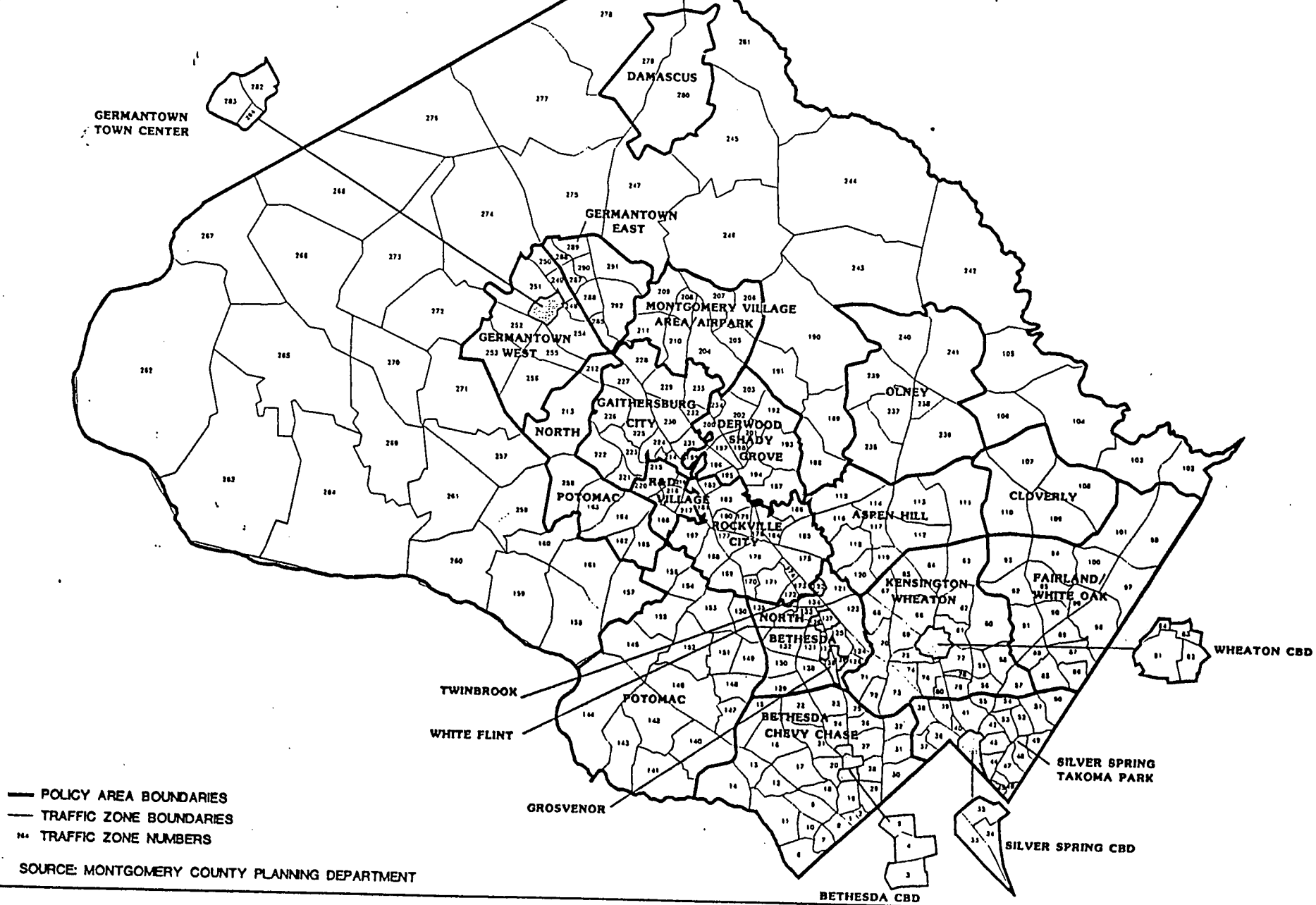


Figure 1: Policy Areas in Montgomery County



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